A-level

# ENVIRONMENTAL SCIENCE

## Paper 2 - Generated practice paper Set 38

## Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

### Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance | Marks |
|---|---|---|
| **01.1** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Zooxanthellae are photosynthetic algae living inside polyp cells (up to 30 000 per $mm^2$)<br>**1** They use the polyp's $CO_2$ and excretory wastes (nitrogen, phosphorus) for photosynthesis and growth, so nutrients are recycled efficiently<br>**1** They supply the polyp with glucose and oxygen - over 95% of its food energy<br>– the polyp provides a protected habitat and nutrients<br>**1** Polyps also trap plankton with cilia and stinging nematocysts<br>– surplus $CO_2$ is used to build the calcium carbonate skeleton<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 4 |



Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **01.2** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** High light (shallow, clear water) for the algae<br>– warm stable temperatures (about 18-30 °C)<br>– low turbidity<br>– constant salinity<br>– well-oxygenated water<br>**1** Found only where all these coincide (about 1% of ocean area)<br>**1** Small changes (warming, sediment, freshwater runoff) move conditions outside the narrow range of tolerance<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **abiotic** → non-living / physical factors<br>• **range of tolerance** → the range of conditions in which a species can survive | 3 |
| **01.3** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1 Fisheries:** support over a quarter of marine species including fish, lobsters, conch<br>**1 Erosion protection:** complex structure dissipates wave energy protecting low coral islands<br>– coral sand builds shores<br>**1 Medicinal discoveries:** AZT and Ara-C from sponges, dolastatin from sea hares, bryostatin from Bugula, trabectedin from sea squirts, secosteroids, coral for bone grafts<br>**1 Climate control:** carbonate skeletons sequester CO2<br>**1** Tourism (diving, snorkelling)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks. | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 57-60; AQA spec 3.1.2.3.4. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **02.1** | **2 marks per point**<br>2 marks for each of the 4 points below; award 1 mark where a point is only partly made.<br>**2** Carnivores fed pellets of sandeels, herring, anchovy<br>– overfishing sandeels reduces puffins and terns<br>– reduced by more plant protein / vegetable oil in pellets or farming herbivores (tilapia)<br>**2** High density spreads lice (which spread to wild fish) and infections<br>– pesticides (organophosphates, pyrethroids, hydrogen peroxide) kill wildlife<br>– antibiotics breed resistant bacteria<br>– reduced by lower densities, single-age tanks with flow from young to old, wrasse as biological control, warm-water washes, circulating flow<br>**2** Escaped farmed fish breed with wild fish introducing disadvantageous genes (bright colour), or non-indigenous escapees colonise as predators / competitors<br>– reduced by better cage design, sterile triploid fish (pressure-treated eggs), species that cannot breed locally (Japanese oysters in UK)<br>**2** Faeces and uneaten food decompose causing deoxygenation under cages<br>– reduced by monitored feeding, siting cages in strong currents, effluent treatment / reed beds for freshwater farms, IMTA with seaweed and shellfish absorbing wastes<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **biological control** → using a natural predator, parasite or pathogen of the pest<br>• **deoxygenation** → loss of dissolved oxygen from the water<br>**Do not credit**<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on | 8 |
| **02.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Predators are killed under licence to protect stock, reducing wild populations of protected / valued species<br>**1** Better cage designs, netting, fencing, bird scarers exclude predators without killing<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 356-359; AQA spec 3.5.2.3. | |
| **03.1** | 1 mark for the point below.<br>**1** Cross-bred offspring are healthier / more productive because the problems caused by homozygous recessive genes in inbred varieties are reduced (heterosis)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **03.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Zebu tolerate heat but have a low milk yield<br>– Ayrshire have a high milk yield but are not heat tolerant<br>**1** Cross-breeding combined the desirable characteristics: heat tolerance with a high milk yield<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **tolerance** → the range of conditions in which a species can survive | 2 |
| **03.3** | **Any 2 of 3**<br>1 mark each for any 2 of the 3 points below, in any order, maximum 2.<br>**1** Replacement of valuable animals that die with genetically identical individuals<br>**1** Replacement of herds culled during a disease outbreak<br>**1** Production of large numbers of individuals with desirable characteristics<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 321-322; AQA spec 3.5.1.2. | |
| **04.1** | **2 marks per point**<br>2 marks for each of the 3 points below; award 1 mark where a point is only partly made.<br>**2 Image databases:** identifying individuals (tiger stripes, whale shark spots, dolphin fins) to find territory size, movements, lifespan, social groups - few species have unique recognisable features<br>**2** eDNA: detecting the presence of a species from DNA shed into water (great crested newt, invasive carp) - shows presence not abundance<br>– contamination / DNA transport by water<br>**2 Geolocator tags:** tracking small birds on migration where transmitters are too heavy - only record light and time so position is approximate<br>– animal must be recaptured to download data<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 6 |
| **04.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Microwaves emitted by the satellite reflect off the surface giving data on wave height, wind velocity, sea surface altitude and oil slicks (which smooth the surface)<br>**1** Microwaves pass through cloud so monitoring is not interrupted by weather<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **04.3** | **Any 2 of 3**<br>1 mark each for any 2 of the 3 points below, in any order, maximum 2.<br>**1 Droppings:** diet, gender, territory size<br>**1 Owl pellets:** diet<br>**1** Feeding marks (chewed hazel nuts - dormouse), tracks / footprints, nests / burrows, scratching posts (territory)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402-407; AQA spec 3.7.2.3. | |
| **05.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1 South America:** high loss (5.1-5.5 Mha $yr^{-1}$) falling by about half by 2020<br>**1 Africa:** loss steady at about 3.4 then rising to 3.9 by 2020 - now the highest<br>**1 Europe:** negative values - forest area increasing (afforestation exceeds loss), though the gain is shrinking<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **05.2** | **Marks shared**<br>Up to 4 marks in total - share them between the 3 points below according to how completely each one is made.<br>**up to 2 South America:** clearance for commercial cash crops (sugar cane, soya) and cattle ranching for export<br>– HEP reservoirs<br>– roads improving access<br>– mining<br>**up to 2 Africa:** growing rural populations clearing land for subsistence farming<br>– fuelwood and charcoal collection above regrowth (charcoal wastes much of the wood's energy)<br>– fodder cutting in dry areas<br>**up to 2 Both:** hardwood timber (teak, mahogany) from clear-felled mixed forest often not replanted<br>– urban expansion<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **HEP** → electricity generated from falling water<br>**Do not credit**<br>• a dam described as having no environmental impacts - flooded land, blocked fish migration, trapped sediment and methane from drowned vegetation | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **05.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** They create barriers fragmenting wildlife populations<br>**1** Improved access makes it easier to remove timber and farm produce, so clearance spreads along them<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **05.4** | 1 mark for the point below.<br>**1** Conifers grow quickly so plantations are replanted after harvest and yield a saleable crop within decades<br>– slow-growing hardwoods (teak, mahogany) take too long for commercial plantations so they are taken from natural forest that is clear-felled and not replanted<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 1 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 367-368; AQA spec 3.5.3.3. | |
| **06.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1 In-situ:** conserving species in their natural habitat with the whole community<br>– ex-situ: away from the natural habitat, eg zoos, botanic gardens, seed banks, captive breeding<br>**1** Needed when conservation in the wild is impossible or insufficient to prevent extinction (habitat gone, population too small, poaching)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• The converse of any point, ie the same comparison written the other way round.<br>• **captive breeding** → breeding in captivity to increase numbers for release<br>**Do not credit**<br>• captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **06.2** | **Any 4 of 5**<br>1 mark each for any 4 of the 5 points below, in any order, maximum 4.<br>**1** Is the wild population threatened?<br>**1** Is there a genetically diverse captive population?<br>**1** Is in-situ conservation failing?<br>**1** Is keeping a captive population realistic (habitat size, food, relationships, cost)?<br>**1** Is release likely to be successful now or in future (suitable habitat, threats removed)?<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **captive breeding** → breeding in captivity to increase numbers for release<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.<br>• captive breeding alone as conservation - credit needs reintroduction, habitat protection or genetic management as well | 4 |
| **06.3** | **2 marks per point**<br>2 marks for each of the 2 points below; award 1 mark where a point is only partly made.<br>**2 Blue whale:** habitat size far too large to provide<br>– food (huge quantities of krill / plankton) cannot be supplied<br>– cost<br>**2 Large blue butterfly:** complex inter-species relationship - larvae are taken into ant nests where they overwinter feeding on ant grubs<br>– requires specific ant species and thyme food plant<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 4 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 32-34; AQA spec 3.1.2.3.3. | |
| **07.1** | **Correct answer only**<br>1 mark for the correct option. No mark if more than one box is ticked.<br>**1 C:** Waste products that are the raw materials for other processes | 1 |
| **07.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Bacteria fix nitrogen at low temperature using enzymes and energy ultimately from sunlight (renewable)<br>**1** The Haber process needs high temperature and pressure powered by non-renewable fossil fuels, releasing $CO_2$<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **nitrogen fixation** → nitrogen gas converted into ammonium or other nitrogen compounds<br>• **sustainable** → meeting present needs without stopping future generations meeting theirs<br>**Do not credit**<br>• 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **07.3** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Incineration uses high temperatures / fuel and releases pollutants<br>– nutrients are not returned as usable raw materials<br>**1** Decomposition at low temperature recycles nutrients as the raw materials for other natural processes (material cycle)<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **incineration** → burning waste, usually with energy recovery<br>• **sustainable** → meeting present needs without stopping future generations meeting theirs<br>**Do not credit**<br>• 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future | 2 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 373; AQA spec 3.6.2-3.6.3. | |
| **08.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Negative feedback responds to any change by re-establishing the previous equilibrium<br>**1** Only large unusual events overwhelm the feedback so a new equilibrium forms<br>**1 Example:** methane hydrate release at the end of the Permian causing climate change and mass extinction<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any appropriate named example, including ones that are not listed here.<br>• **negative feedback** → a change that cancels itself out / a self-correcting cycle | 3 |
| **08.2** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** In a diverse system each component is relatively less important so the loss / change of one has a small effect<br>– alternative food sources and relationships buffer the change<br>**1 Natural:** coral reefs and tropical rainforests with complex food webs are stable<br>– simple polar / desert systems fluctuate<br>**1 Human:** agroecosystems focus on a few crops and remove detritivores, pollinators and predators so are vulnerable to pests, disease, weather<br>**1 Diverse technical systems:** a society using a range of renewable energy resources is more secure than one relying on a single source<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **08.3** | **Any 3 of 7**<br>1 mark each for any 3 of the 7 points below, in any order, maximum 3.<br>**1** Melting permafrost releasing methane<br>**1** Reduced albedo as ice / snow melt<br>**1** Methane hydrate release<br>**1** Forest and peat fires<br>**1** Faster soil decomposition<br>**1** Ocean acidification reducing sequestration<br>**1** Cirrus cloud formation<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• **positive feedback** → a change that makes itself larger / a self-reinforcing cycle<br>• **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• carbon dioxide as the main cause of acid rain - the gases responsible are sulfur dioxide and oxides of nitrogen<br>• 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 371-372; AQA spec 3.6.1. | |
| **09.1** | **1 mark each**<br>1 mark for each of the 3 points below.<br>**1** Other MEDCs (USA, Canada, Australia) also had surpluses<br>– Japan already had suppliers<br>**1** Eastern Europe / USSR needed food but could not pay<br>**1** Selling cheaply in LEDCs would undercut local farmers, put them out of business and reduce their long-term food production<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 3 |
| **09.2** | **1 mark each**<br>1 mark for each of the 4 points below.<br>**1** Quotas limiting production (eg milk)<br>**1** Set-aside payments for taking land out of production<br>**1** Farm diversification (recreation, farm shops, specialised products) and alternative crops / livestock (biofuels, pharmaceutical poppies, llamas, deer, bees)<br>**1** Agri-environmental schemes (ESA, CSS, ESS, CS) paying for environmentally beneficial farming<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first four - the question asks for four, so a list of extras cannot pick up the marks. | 4 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **09.3** | **Any 3 of 5**<br>1 mark each for any 3 of the 5 points below, in any order, maximum 3.<br>**1** Hedgerows at least 1.5 m tall cut at 2-year intervals outside the breeding season - bird / insect habitat and landscape<br>**1** Buffer strips / field margins (no cultivation, fertiliser or pesticide within 2 m) - protect ditches and rivers, habitat<br>**1** Beetle banks of tussocky grass - natural pest predators<br>**1** Skylark nesting plots<br>– wild bird seed / nectar plants - winter food and pollinators<br>**1** In-field tree protection<br>– controlled stocking density to reduce compaction<br>– ditch management<br>– dry-stone walls<br>– woodland management<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>**Do not credit**<br>• Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.<br>• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on | 3 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 332-334; AQA spec 3.5.1.4. | |
| **10.1** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Plants produce chemicals (often alkaloids) that are toxic to the herbivores that would eat them<br>**1** In carefully controlled amounts these chemicals have beneficial physiological effects in humans<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |
| **10.2** | **1 mark each**<br>1 mark for each of the 2 points below.<br>**1** Only a very small proportion of species have been studied for medicinal substances, so it is not known which species will be useful<br>**1** Conserving the habitat protects all the species in it, including those not yet studied and the species they depend on<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes. | 2 |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| **10.3** | **Any 2 of 3**<br>1 mark each for any 2 of the 3 points below, in any order, maximum 2.<br>**1** Marsupials (kangaroo / wallaby): young develop in the pouch so are easier to study than a human foetus - developmental problems<br>**1 Hippopotamus:** hipposudoric acid skin secretion (sunscreen / antimicrobial) - burns treatment<br>**1 Marine sponges:** proteins preventing graft rejection - organ transplants<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any appropriate named example, including ones that are not listed here. | 2 |
| **10.4** | **Levels of response**<br>Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist.<br>**Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached.<br>**Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported.<br>**Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation.<br>**0 marks**: No relevant content.<br>**Indicative content - a guide to placing the mark, not a checklist**<br>• **Resources:** wood, fibres, oils, fuels, food<br>– new food species adapted to local conditions (morama bean, yeheb, perennial maize, Kernza, bison, eland)<br>• **Biomimetics:** whale flipper tubercles, shark skin, termite-mound ventilation, bird bones, gecko adhesion, burrs / Velcro, lotus self-cleaning, spider silk<br>• Medicines and physiological research<br>– genetic resources - crop wild relatives, Vavilov centres, gene pools, inbreeding<br>• **Ecosystem services:** atmospheric composition, hydrological cycle, biogeochemical cycles, soil maintenance<br>– interspecies relationships - food, pollination, seed dispersal, habitat provision<br>• **Evaluation:** services go unnoticed because of dynamic equilibrium<br>– unknown future value<br>– economic vs moral / intrinsic arguments<br>– conservation also needed for aesthetic and cultural reasons<br>• Justified conclusion<br>**Also accept**<br>• Any other correct, relevant point of the same kind - wording never has to match these notes.<br>• Any accurate, relevant material that is not in the indicative content.<br>• **biodiversity** → the variety of life: habitat, species and genetic diversity | 9 |
| | **Model answer - Level 3 (9 marks)** | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| | Other species supply resources we use directly: wood, fibres such as cotton and wool, oils, fuels and food. Because most farmed species were domesticated long ago and are often kept where they are poorly adapted, wild species such as the drought-resistant morama bean, the yeheb tree that grows on poor soils, perennial maize that needs no ploughing, and bison that graze grassland too poor for cattle offer new, better-adapted food sources. Wild species also carry genetic resources: crop wild relatives in Vavilov centres hold genes for disease resistance and salt tolerance that inbred commercial varieties lack. | |
| | Biomimetics copies adaptations into technology - humpback flipper tubercles in turbine blades, shark skin in friction-reducing coatings, termite-mound convection in naturally ventilated buildings, hollow bird bones in lightweight roofs. Medicines such as taxol, quinine, AZT and diosgenin came from wild species, and physiological research on marsupials, hippo skin and sponge proteins is helping with foetal development, burns and transplant rejection. Since only a tiny fraction of species has been studied, the unknown future value argues for conserving whole habitats. | |
| | Beyond resources, other species provide services we could not replace: regulation of atmospheric gases, evapotranspiration driving rainfall, the carbon, nitrogen and phosphorus cycles run by microbes, soil formation by decomposers, and pollination, seed dispersal and habitat provision. These are unnoticed precisely because they are in dynamic equilibrium, so their economic value is underestimated. On balance the resource and service arguments alone justify conservation, but they are strongest when combined with the intrinsic and aesthetic case, because many species with no known use still sustain the systems we depend on. | |
| | **Weaker answer - Level 1 (3 marks)** | |
| | Biodiversity is important because we get lots of things from other species. We get wood, cotton and food from plants and animals. Some medicines come from plants, like aspirin from willow trees. Animals can help design things, like Velcro was copied from seeds. Other species also do things for us like bees pollinating crops. If we lose species we might lose things that could be useful in the future. So we should conserve biodiversity. | |
| | **What makes the difference** | |
| | • Both answers cover resources, medicines, biomimetics and services, but Level 3 develops each with specific, accurate detail (which species, what property, why it matters). | |
| | • The question asks 'to what extent'. Level 3 weighs resource arguments against intrinsic value and reaches a reasoned conclusion; Level 1 asserts 'so we should conserve'. | |
| | • Level 3 explains *why* services are undervalued (dynamic equilibrium) - an explanation, not just a list. | |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-19; AQA spec 3.1.2.1. | |

Do not write outside the box

| Question | Marking guidance | Marks |
|---|---|---|
| 11.1 | **Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• Sustainability definitions<br>– population and per capita impact<br>– attitudes of rural and urban societies<br>• **Negative feedback:** cloud cover and albedo, $CO_2$ and photosynthesis, evaporation and precipitation, density-dependent population regulation<br>• Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition<br>• **Tipping points:** rare under negative feedback<br>– large events (Permian methane hydrate)<br>– self-sustaining change from human activities<br>• **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators<br>• Human responses that override feedback (fertilisers, irrigation, pesticides)<br>– links to climate change chapter<br>– justified conclusion<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* | 25 |
| | **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1. | |

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| Question | Marking guidance | Marks |
|---|---|---|
| 11.2 | **Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.**<br><br>**Levels of response**<br>**Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached.<br>**Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified.<br>**Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics.<br>**Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions.<br>**Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate.<br>**0 marks**: No relevant content.<br><br>**Indicative content - students are not expected to cover all of these**<br>• **Temperate broadleaf woodland:** seasonality, deep soils, high biodiversity, resources, climate and soil regulation<br>– deforestation, fragmentation, management change<br>– SSSIs, ancient woodland protection, coppicing, mixed planting<br>• **Tropical rainforest:** stable warm wet climate, stratification, very high biodiversity, carbon and hydrological roles, thin soils<br>– fuelwood, timber, agriculture, mining, reservoirs, climate change<br>– protected areas, debt-for-nature swaps, sustainable exploitation<br>• **Tropical coral reefs:** cnidarian-zooxanthellae symbiosis, light, temperature, turbidity, salinity<br>– fisheries, erosion protection, medicines, tourism<br>– damage, souvenirs, sedimentation, pollution, bleaching, fishing, introduced species<br>– MPAs, no-take zones, control of activities<br>• **Deep-water reefs:** cold, dark, slow growth<br>– research, fisheries<br>– trawling, oil / gas, acidification<br>– SACs<br>• **Oceanic islands:** isolation, endemism, no mammal predators | 25 |

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| Question | Marking guidance | Marks |
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| | – exploitation, introduced species, habitat change, sea level<br>– eradications, control of development and visitors<br>• **Mangroves:** halophytes, anaerobic mud<br>– erosion protection, fisheries nurseries, sediment trapping<br>– clearance for aquaculture / urban / ports, pollution, climate change<br>– protection, replanting<br>• **Antarctica:** extreme cold, seasonality, marine nutrients, ice<br>– water store, albedo, carbon, research<br>– climate change, ozone, tourism, fishing, minerals<br>– Antarctic Treaty, CCAMLR, tourism and waste rules<br>• **Synthesis:** features -> importance -> threats -> strategy<br>– evaluation of success<br>– conclusion<br>*Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*<br>**Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-71; AQA spec 3.1.2.3.4. | |

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